Network Science Lab, Week 3

Analyzing Ego Networks and Community Structure in Social Media

Objective: This lab guides you through a complete network analysis workflow on a real-world social network. You will use Python for data loading, preprocessing, and large-scale computation, and Cytoscape for deep visualization, exploration, and validation of your results.

Dataset: Facebook Ego Network (from SNAP).

- Nodes: ~4,039 users from a subset of the Facebook social graph.
- Edges: ~88,234 undirected friendships between these users.

Prerequisites:

- Python 3.x with modules: networkx, pandas, requests, python-louvain
- Cytoscape 3.9+

Part 1: Data Acquisition and Preprocessing in Python

1. Download and Load the Network:

- HINT: Use requests.get() to fetch the data and gzip.decompress() to handle the compression. Create a file-like object with BytesIO.

- FUNCTION: Use read_edgelist() to load the data into an undirected graph (G).
- FUNCTION: Use G.number_of_nodes() and G.number_of_edges() to print the graph's basic information.

2. Initial Interrogation & Isolating the Largest Component:

- Check connectivity with is_connected().
- Find all components with connected_components(). Use max() with key=len to find the largest one. (LCC)
- Create a subgraph of the LCC using G.subgraph(). Use .copy() to create a new graph object.
- Print the size of the LCC using the same methods as in step 1.

Deliverable 1:

- Report the sizes of the original graph and its LCC.
- What does the existence of multiple components suggest about a social network?

Part 2: Advanced Centrality and Preparing for Cytoscape

3. Compute Key Centrality Metrics:

- FUNCTIONS: degree_centrality, pagerank, set_node_attributes betweenness_centrality

- Use set_node_attributes() to add each centrality result as a new attribute to the graph

4. Export for Cytoscape:

- Use write_graphml to export the graph with all its attributes.

Deliverable 2:

- Why is the k parameter used in the betweenness centrality calculation? What is the trade-off?

Part 3: Visualization and Exploration in Cytoscape

5. Visualize the Network:

- In Cytoscape: File -> Import -> Network from File...
- Apply a layout: Layout -> Prefuse Force Directed Layout

6. Style the Network:

- In the Style panel:
  - Map Node Size to pagerank (Continuous Mapping)
  - Map Node Color to betweenness_centrality (Continuous Mapping)

Deliverable 3:

- Include a screenshot of your styled network.
- Based on the visualization, are the most central nodes scattered or clustered? What does this suggest?

7. Identify Top Influencers:

- Use the Table Panel at the bottom of Cytoscape.
- Sort the degree_centrality column in descending order.
- Select the top 10 rows, right-click, and choose Select Nodes from Selected Rows.

Deliverable 4:

- Create a table showing the node ID and its rank for degree, PageRank, and betweenness for the top 5 nodes.
- Are the same nodes consistently at the top? If not, explain the difference.

Part 4: Community Detection and Validation

8. Detect Communities in Python:

- IMPORT: from community import community_louvain
- Use community_louvain.best_partition
- Use set_node_attributes() to add the resulting partition dictionary as a new node attribute (e.g., 'louvain_community').
- Re-export the graph with write_graphml().

9. Analyze Communities in Cytoscape:

- Import the new GraphML file or update the table
- In the Style panel, change Node Fill Color to map to louvain_community (Discrete Mapping).

Deliverable 5:

- How many communities were detected?
- Does the visual mapping of communities correspond well with the network layout?

Part 5: Robustness and "Attack" Simulation (Python)

10. Simulate Network Attacks:

- Use sorted(centrality_dict.items(), key=lambda x: x[1], reverse=True)[:n] to get the top n nodes by a centrality measure.
- Use G.copy() to create a working copy of the graph.
- Use graph_copy.remove_nodes_from(top_nodes_list) to perform the attack.
- Use max(connected_components(graph_copy), key=len) to find the new LCC after attack.
- Run this simulation for both degree and betweenness centrality.

Deliverable 6:

- Report the proportion of the network remaining in the LCC after each attack.
- Which attack strategy was more effective? What does this tell you about the network's vulnerability?

Part 6: Synthesis and Proposal

11. Strategic Recommendation:

- a) Viral Marketing: Which 5 users would you target? Justify using your centrality analysis.
- b) Risk Mitigation: Which user type (hubs or bridges) poses a greater risk? Propose mitigation strategies.

12. Reflection:

- What were the benefits of using Python? What were the benefits of using Cytoscape?
- Propose another real-world application for this approach. Define nodes, edges, and the most critical metric.